Documentation for Sonar Data Classification Project

Project Overview

This project implements a logistic regression model to classify sonar signals as either rocks or mines. The model achieves high accuracy in distinguishing between these two classes, which has applications in underwater object detection and security systems.

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Project Structure

The notebook follows a standard machine learning workflow:

1. Import dependencies
2. Data collection and processing
3. Data splitting
4. Model training
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6. Prediction system

Dependencies

python

```
import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score
```

Data Description

The dataset contains 208 samples with 60 features representing energy measurements at different frequencies

- Target variable (column 60) has two classes:
  - 'M' for Mine
  - 'R' for Rock
- Dataset is well-balanced with similar number of samples for each class

Implementation

Data Processing

- Loaded data using pandas
- Separated features (X) and labels (Y)
- Used stratified train-test split (90-10) to maintain class distribution

Model Training

- Logistic Regression was chosen as the classification algorithm
- Trained on 90% of the data (187 samples)

Evaluation

- Achieved 83.42% accuracy on training data
- Achieved 76.19% accuracy on test data

Results

The model shows good performance in distinguishing between rocks and mines based on sonar signals. The predictive system can take new input data and classify it as either a rock or mine with reasonable accuracy.